

YatraAI

Context-Aware Group Travel Intelligence

A group itinerary planner where a constraint solver — not a language model — decides what you do and when. The model runs last, writes prose, and changes nothing.



THE PROBLEM — AND THE ONE DESIGN DECISION EVERYTHING FOLLOWS FROM

WHAT THE PRODUCT SOLVES

Business problem. Group trips fail on coordination, not inspiration. Four people want four different things; one always gets ignored; the plan ignores opening hours, travel time and weather, and collapses on day one.

Technical problem. Produce a fair, explainable and physically feasible group itinerary under weather, time windows, budget, distance, opening hours, individual preferences and accessibility — without letting a language model invent the facts.

WHY AN LLM CANNOT OWN THIS

Ask any LLM to "plan three days in Agra" and it will confidently:

- schedule the Taj Mahal on a Friday — it is closed to tourists
- allow 20 minutes to cross Delhi in traffic
- quote a ticket price it invented

Each failure is a constraint violation, not a wording problem.

RESPONSIBILITY SEPARATION — WHO IS ALLOWED TO INVENT ANYTHING

STAGE	OWNER	CAN IT INVENT FACTS?
1	Curated catalogue + provenance	No — every field cites a source
2	Hard eligibility filters	No — infeasible options removed
3	Fairness-aware ranking (9 features)	No — deterministic
4	Balanced k-means day clustering	No
5	OSRM → haversine travel matrix	No
6	OR-Tools CP-SAT scheduler	No — solves a stated model
7	15-check constraint validator	No — errors block display
8	LLM (optional)	YES — which is why it runs last

The load-bearing claim. The LLM sits at stage 8. By then the itinerary is chosen, scheduled and validated. Its output is prose printed beside the timeline — it cannot add, remove or reorder a stop.

Proof, not assertion. The default provider is mock. Live telemetry at /api/v1/metrics reports `llm.calls = 0`. Every itinerary in this deck was produced with no model call. Switch a real provider on and the plan is byte-identical — only the paragraph changes.

SO WHAT? Correctness is delegated to systems that can be tested; fluency is delegated to a model that cannot break correctness. That split is the whole architecture.

SYSTEM ARCHITECTURE & TECHNOLOGY STACK

REQUEST PATH — GENERATE AN ITINERARY

1

Next.js 14 client

typed API client, TanStack Query cache

2

FastAPI route

Pydantic validation · JWT · rate limit tier

3

Planner pipeline

8 stages, pure-Python domain objects

4

OR-Tools CP-SAT

one model per day, deterministic

5

Validator + persist

15 checks, then SQLAlchemy write

STACK BY LAYER	
LAYER	TECHNOLOGY
Frontend	Next.js 14 (App Router) · TypeScript strict · Tailwind · TanStack Query · Recharts · react-leaflet
API	FastAPI · Pydantic v2 · pydantic-settings · structlog
Persistence	SQLAlchemy 2.0 · Alembic · PostgreSQL + pgvector, portable to SQLite
Optimisation	Google OR-Tools CP-SAT
ML / science	NumPy · pandas · scikit-learn · Pandera
Data pipeline	Bronze/Silver/Gold medallion · Airflow DAGs · pyarrow
Security	PyJWT (HS256) · bcrypt · in-process sliding-window limiter
Testing / CI	pytest (295) · Vitest + Testing Library (43) · ruff · mypy · GitHub Actions
Deploy	Docker multi-stage · Compose · render.yaml · Vercel

PROVIDER ABSTRACTION — EVERY EXTERNAL DEPENDENCY IS SWAPPABLE, AND EVERY FALLBACK IS MEASURED				
PROVIDER	DEFAULT	FALLBACK	LIVE RATE	WHY IT MATTERS
LLM	mock	deterministic templates	0 calls	Product is fully functional with no model configured — the default the tests run against
Embeddings	hashing	n/a (is the default)	0% fallback	384-dim signed feature hashing — deterministic, no model download, not a neural encoder
Routing	OSRM	haversine × detour factor	100% fallback	Conservative Indian urban speeds; underestimating travel is the dangerous direction
Weather	Open-Meteo	committed climatology	100% fallback	Labelled in the UI as seasonal averages, never presented as a forecast
Database	PostgreSQL + pgvector	SQLite	—	Four portable TypeDecorators (GUID, JSONType, VectorType, TZDateTime) — one schema, two engines

KEY POINT: No provider raises on failure — each degrades to a documented fallback and records it. The UI shows an honest banner; /metrics reports the real fallback rate. Both amber rates above are this machine being offline, not a bug.

DATA FOUNDATION — WHERE THE FACTS COME FROM, AND HOW THEY ARE POLICED

BRONZE → SILVER → GOLD, WITH A CONTRACT AT EVERY PROMOTION

BRONZE	SILVER	GOLD
Raw seed JSON, content-hashed 140 documents ingested	Validated, typed, deduplicated 130 attractions · 142 sources 164 schedules · 216 fee rows	Derived planning features 130 feature rows · 10 cluster metric rows · 0 rejected

WHAT EVERY ATTRACTION CARRIES — ~30 STRUCTURED FIELDS	
GROUP	FIELDS
Identity	slug · name · locality · city · lat/lon · categories (mapped to a 10-interest taxonomy)
Narrative	summary · history · significance · 2–8 interesting facts → these become the RAG chunks
Planning	typical / min / max duration · best time of day · suitable months · indoor-outdoor · weather sensitivity · crowd profile
Access	wheelchair · accessibility notes · senior & child suitability · physical intensity
Etiquette	dress code · photography policy · local customs (mandatory for religious sites — enforced by test)
Schedule	per-weekday open/close windows · seasonal variants · closure days
Cost	fee bands by visitor type · free flags
Provenance	≥1 cited source, HTTPS, allow-listed domain, with a covers[] list and last_verified date

TWO REAL DEFECTS THE GATE CAUGHT: a neighbour reference crossing destination clusters (Mawlynnong → Dawki), and two attractions sharing identical coordinates (Gaurikund / Kedarnath trek). Both were my authoring errors. Neither was visible by reading the JSON.

THE GATES

7 Pandera contracts enforced at promotion — a row that fails is rejected with a reason, never coerced silently.

12 runtime data-quality checks power an admin dashboard and gate the Airflow DAG.

30 data-quality tests run in CI: coordinate bounds, cluster distance, duplicate detection, category-taxonomy coverage, source allow-list, dress code on religious sites.

THE HONESTY RULE

Not one schedule or fee row in this dataset claims to be verified.

I authored the catalogue from public sources; I cannot verify opening hours or prices. So every such row carries verified: false, the UI renders a badge, and a data-quality test fails the build if a row ever claims otherwise.

RANKING & GROUP FAIRNESS — TURNING FOUR PEOPLE'S PREFERENCES INTO ONE SHORTLIST

STEP 1 — HARD ELIGIBILITY (REMOVAL, NOT DOWN-RANKING)

Closed on every day of the trip · outside budget · fails a stated accessibility need · on a member's avoid list.

A member veto is absolute — a majority cannot outvote one person's mobility requirement. Infeasible options are deleted from the candidate set, so the optimiser can never trade an accessibility need against a higher score.

STEP 2 — NINE EXPLAINABLE SCORE COMPONENTS	
COMPONENT	WHAT IT MEASURES
interest_match	Member interests vs the attraction's mapped categories
group_fairness	Marginal gain to the least-served members
attraction_quality	Curated significance / interest
seasonal_suitability	Month fit, per attraction
weather_suitability	Forecast vs weather sensitivity + indoor/outdoor
budget_suitability	Entry fee against the daily budget
accessibility_suitability	Access rating and physical intensity
evidence_quality	Source authority, count and completeness
distance / crowd penalty	Subtracted — travel cost and congestion

Weights are fixed and inspectable. The "Why was this selected?" panel renders the stored breakdown for that plan — it is not generated text, and it cannot disagree with the optimiser.

INTERVIEW LINE: "I did not pick an aggregation rule, I measured five. Borda scored the best mean and the worst floor — which is the trap. The shipped selector deliberately gives up 1.1% of mean satisfaction to raise the least-satisfied member 8.5%."

STEP 3 — AGGREGATION: FIVE METHODS, MEASURED HEAD-TO-HEAD

METHOD	MEAN SAT.	LEAST SATISFIED	SPREAD
simple_average	0.604	0.481	0.172
borda_count	0.618	0.478	0.197
max_min_fairness	0.582	0.545	0.069
fairness_aware (1-pass)	0.603	0.484	0.167
select_fairly (SHIPPED)	0.597	0.522	0.115

How to read this. Five group profiles built so a simple average steamrolls a minority. Iterative fair selection captures 64% of pure max-min's fairness gain for 30% of its cost in mean satisfaction.

The null result is the useful one: single-pass fairness_aware barely beats a plain average (0.484 vs 0.481). That is exactly why the iterative selector exists — the cheap version does not work.

THE FAIRNESS METRIC

Jain's fairness index $J(x) = (\sum x_i)^2 / (n \cdot \sum x_i^2)$

1.0 = everyone equally served; 1/n = one person served. Reported alongside the floor guaranteed to the worst-served member, because an average hides exactly the person this product exists to protect.

THE OPTIMISER — A PRIZE-COLLECTING TSP WITH TIME WINDOWS, SOLVED PER DAY

THE CP-SAT MODEL

DECISION VARIABLES

$x[i][j] \in \{0,1\}$ tour goes $i \rightarrow j$ $x[i][i] = 1$ node skipped
 $u[i] \in \mathbb{Z}$ service start, minutes from midnight

CONSTRAINTS

C1

AddCircuit over all arcs — sequencing AND selection in one constraint. A node opts out via its self-loop: no subtour-elimination family, no big-M.

C2

$x[i][j] = 1 \Rightarrow u[j] \geq u[i] + s_i + t_{ij}$ (a real implication, via OnlyEnforceIf)

C3

$e_i \leq u[i] \leq l_i$ — the weekday opening window $\cap$ the group's day window

C4

skipped nodes pinned to e_i , so unvisited starts cannot float and slow the solve

C5

must-visit places and the meal break are mandatory nodes

C6

caps on activities, daily travel minutes and entry-fee budget

OBJECTIVE

maximise $\sum [(p_i - \text{shift} + 0.05) \cdot 1000] \cdot v[i] - 4 \cdot \sum t_{i,j} \cdot x[i][j]$

THREE DECISIONS AN INTERVIEWER WILL PROBE

Why per-day, not trip-wide?

Trip-wide is $O((k\text{-days})^2)$ arcs — 1,296 vs 324 for $k=9$, 4 days. It also zig-zags, because a global objective only sees total distance.

Why the score shift?

Scores can go negative after penalties; a negative prize makes skipping rewarding and the solver returns an empty day.

Why TRAVEL_WEIGHT = 4?

It is what stops two high-scoring places at opposite ends of a city beating three good ones in a cluster. The single most consequential constant in the model.

MEASURED — OR-TOOLS vs A FAIR GREEDY BASELINE

METRIC	GREEDY	CP-SAT	Δ
Total travel distance	149.8 km	64.0 km	−57%
Total travel time	487 min	275 min	−43%
Travel km per activity	15.61	6.01	−61%
Activities scheduled	9.8	10.3	+0.5
Constraint violations	0.0	0.0	—
Median solve time	8.8 ms	48.2 ms	+39 ms

The baseline is not a strawman.

It honours identical opening hours, budget, activity and travel caps, and inserts each candidate at its earliest feasible start. The 57% gap measures sequencing quality alone.

RELAXATION LADDER — WHY IT NEVER JUST SAYS "NO PLAN"

TRY	WHAT IS RELAXED
1	Nothing — the full model
2	max_activities − 1
3	travel cap × 1.25
4	drop budget cap; activities − 1; travel × 1.4
5	activities = 2; travel × 1.5

Travel is relaxed last and least — a six-hour driving day is feasible and useless. Whatever was relaxed is recorded and shown to the user.

DETERMINISM: num_workers = 1 and a fixed random seed. Multi-threaded CP-SAT search is non-deterministic; single-worker costs some speed and buys a reproducible plan — asserted by test.

VALIDATION & DYNAMIC REPLANNING — WHY A WRONG PLAN CANNOT REACH THE SCREEN

THE VALIDATOR — 15 CHECKS, RE-DERIVED INDEPENDENTLY OF THE SOLVER	
CHECK	SEVERITY
Activities ordered and non-overlapping	error
Every visit inside a real opening window for that weekday	error
Nothing before day start or after day end	error
Visit duration within the attraction's min/max	error
Travel time between consecutive stops is actually available	error
No attraction scheduled twice across the trip	error
Every mandatory attraction is scheduled	error
Accessibility requirements respected	error
Nothing scheduled on a day the attraction is closed	error
Daily travel / activities within the pace guideline	warning
Meal break exists on any day longer than 6 h	warning
Total estimated cost within budget	warning

Independent by design. The validator re-reads the persisted rows and re-derives every constraint from scratch. A modelling bug therefore surfaces as a failed validation, not as an impossible itinerary shown to a user.

Budget is a warning, not an error — overspending is information the traveller should act on, not a reason to withhold the plan.

ELEVEN REPLANNING ACTIONS — EXPRESSED AS INPUTS, NEVER AS EDITS
remove_activity · replace_activity · regenerate_day · make_day_relaxed · reduce_cost · reduce_travel · add_theme · shift_start_time · weather_replan · member_opted_out · attraction_unavailable Nothing edits a schedule in place. Each action changes planner inputs, then the whole pipeline re-runs and re-validates. A modification therefore cannot produce a state the optimiser would never have generated.

GROUP CONTROL — A CHANGE IS A PROPOSAL, NOT A FAIT ACCOMPLI
On a trip with more than one active member, a modification becomes a ChangeProposal with an approval threshold. Members vote; reaching the threshold auto-applies it. The owner can override, and the override is written to the append-only audit log. The API returns <code>requires_group_approval: true</code> and a <code>proposal_id</code> rather than silently applying one member's preference.

LIVE SOLVER TRACE — SURFACED IN THE UI, NOT JUST LOGGED				
DAY	STATUS	CHOSEN	OBJECTIVE	SOLVE
1	OPTIMAL	4 / 9	6,393	59.3 ms
2	OPTIMAL	4 / 6	2,468	48.2 ms
3	OPTIMAL	1 / 1	712	3.4 ms

WHAT THIS DEMONSTRATES: the product shows its own working. Per-day CP-SAT status, wall-clock, objective value, candidates considered and any relaxation are rendered on the itinerary page, alongside an input fingerprint proving the same inputs reproduce the same plan.

RETRIEVAL — ANSWERING FROM EVIDENCE, OR REFUSING TO ANSWER AT ALL

THE PIPELINE

1 Classify

Question → knowledge topic (history, etiquette, access, visiting)

2 Filter

Metadata scope by cluster and place — an attraction filter REQUIRES a cluster

3 Retrieve ×2

Dense cosine over 790 chunks + BM25, run independently

4 Fuse

Reciprocal Rank Fusion, k = 60

5 Rerank

Query coverage · heading match · topic agreement · source authority

6 Answer or abstain

Absolute evidence score vs threshold 0.26

TWO MECHANISMS THAT MAKE THE OUTPUT TRUSTWORTHY

Citations are computed, not claimed.

After generation, each answer sentence is matched back to the chunk it overlaps most (threshold 0.30). A sentence the model invented earns no citation. Asking a model to cite itself measures nothing.

Abstention runs on an absolute score.

evidence = max(0.35·dense + 0.65·IDF-weighted term coverage).
Below 0.26 no model is called at all and the answer is "I don't have verified information", with zero citations.

MEASURED — 34-QUESTION COMMITTED BENCHMARK		
METRIC	VALUE	NOTE
Open retrieval top-1	0.903	no attraction hint given
Open retrieval top-3 / MRR	1.000 / 0.952	the realistic assistant path
Retrieval recall	1.000	expected topic retrieved
Topic classification	0.941	was 0.50 — see defect below
Citation correctness / coverage	1.000 / 1.000	derived, not requested
Abstention accuracy	1.000	refuses when it should
Faithfulness	0.995	our disclaimer counts against
Evidence: answerable / out-of-scope	0.582 / 0.071	what abstention keys on
Latency p50 / p95	8.1 / 34.4 ms	template provider

ABLATION — REAL RE-RUNS, NOT RE-SORTS		
CONFIGURATION	TOP-1	MRR
hybrid + rerank (shipped)	0.903	0.952
hybrid, no rerank	0.774	0.874
dense only + rerank	0.871	0.935
lexical only + rerank	0.903	0.946

Reranking is worth +12.9 pp top-1. Lexical-only nearly matches the hybrid — an honest limitation of the hashing embedder (a lexical projection), not a strength of BM25.

THE DEFECT THAT ONLY MEASUREMENT COULD FIND: "What happened at Sarnath and why does it matter?" abstained at 0.229 — with the correct chunk ranked FIRST — because "happened" and "matter" appear in 0 of 790 chunks and so drew maximum IDF weight. Absent entities signal out-of-scope; absent framing words signal nothing. Fixing that moved abstention 0.912 → 1.000 while out-of-scope evidence stayed at exactly 0.071.

SECURITY, PRIVACY & THE HONESTY CONTRACT

SECURITY — THREATS THAT ACTUALLY APPLY, AND THE CONTROL FOR EACH

THREAT	CONTROL
Credential compromise	bcrypt (rounds=12) with a SHA-256 pre-hash — bcrypt silently truncates at 72 bytes
Token forgery	HS256 JWT, signature + issuer + expiry verified; Supabase tokens accepted
Horizontal escalation	require_trip_member / require_trip_owner as dependencies on every trip route
Cost / DoS	Two-tier sliding window — 120 req/min default, 15 req/min on optimiser and RAG
Prompt injection	Detected on the question, neutralised on every retrieved chunk, injection-aware system prompt
Data poisoning	Source-domain allow-list enforced at the Silver gate
Secret leakage	.env git-ignored; .env.example holds names only; production boot guard refuses weak config

PRODUCTION BOOT GUARD

With `YATRA_ENV=production` the API refuses to start if:
JWT_SECRET is under 32 chars or contains 'dev-only' · DATABASE_URL is unset (the SQLite fallback is refused) · any non-localhost CORS origin uses plain http:// · a named LLM provider has no API key.

PRIVACY — EVERY GUARANTEE IS ENFORCED IN CODE AND COVERED BY A TEST

GUARANTEE	IMPLEMENTATION
Opt-in only	No session without consent_granted_at + a stored consent-text version
Approximate by default	Snapped to a ~500 m grid BEFORE storage — the exact point is never written
Always expiring	12 h hard cap per session; every point carries its own expiry
Stop means delete	Stopping deletes the trail immediately, not just hides it
Never in analytics	Coordinate-like keys stripped at write time; a test posts a real coordinate and asserts it appears nowhere
No location history	There is no table that could hold one

THE HONESTY CONTRACT — FOUR CLAIMS THE CODE ENFORCES

- 1 No verified hours or fees.** A data-quality test fails the build if a row ever claims otherwise.
- 2 Cited or silent.** The assistant abstains rather than guessing, and states plainly when a value is recorded rather than live.
- 3 SOS is a demonstration.** It contacts nobody — stated in the OpenAPI description, the response body, the UI, and the chat message it posts, and it refuses to fire without an explicit acknowledgement flag.
- 4 No invented metrics.** Every number in the README re-runs from a script. Synthetic ML results carry data_kind="synthetic" on every row.

WHY THIS MATTERS IN AN INTERVIEW: these are the constraints a real product team would impose, and they are the parts most portfolio projects skip. Being able to say "the build fails if that claim is ever made" is stronger than any feature.

IN ONE SENTENCE

A group itinerary planner where a constraint solver decides, a validator gates, and the language model only ever narrates.

EVERYTHING MEASURED, IN ONE PLACE	
Travel distance vs greedy baseline	-57%
Least-satisfied member vs simple average	+8.5%
RAG abstention accuracy / citation correctness	1.000 / 1.000
Retrieval top-1 · MRR (no hint)	0.903 · 0.952
Reranking contribution to top-1	+12.9 pp
Median CP-SAT solve per day	48.2 ms
Automated tests (295 Python + 43 frontend)	338
Post-deployment verification checks	10 / 10
LLM calls required to produce an itinerary	0

```
Reproduce everything: make experiments · python evaluation/run_rag_eval.py · pytest
```

338 tests passing · 10/10 deployment checks · ruff, mypy, tsc and eslint clean · every figure above regenerated from a script in the repository

STATED LIMITATIONS — LEAD WITH THESE, DO NOT BE CAUGHT BY THEM

Default embeddings are not neural. A 384-dim signed hashing projection. sentence-transformers is a one-env-var swap.

ML models were not promoted. Trained on synthetic labels. The trees lost (0.6404) to the hand-weighted rule (0.6676); the one learner that edged it was inside a bootstrap CI of [-0.010, +0.017].

The benchmark is mine. 34 hand-written questions, not collected from users. Small, and labelled as such.

Ten destinations, not all of India. Depth over breadth — a new cluster is a data change, not a code change.

Rate limiting is per-process; no refresh tokens. Documented in docs/security.md under "known gaps".

FOUR LINES TO SAY OUT LOUD

"It is a retrieval and constraint-optimisation system, not a chatbot. The LLM runs last and changes nothing — the default config never calls it."

"I measured five aggregation rules. The one with the best mean had the worst floor."

"Every interesting bug was found by an experiment, not by reading code."

"No opening hour or fee in this dataset claims to be verified — the build fails if one does."